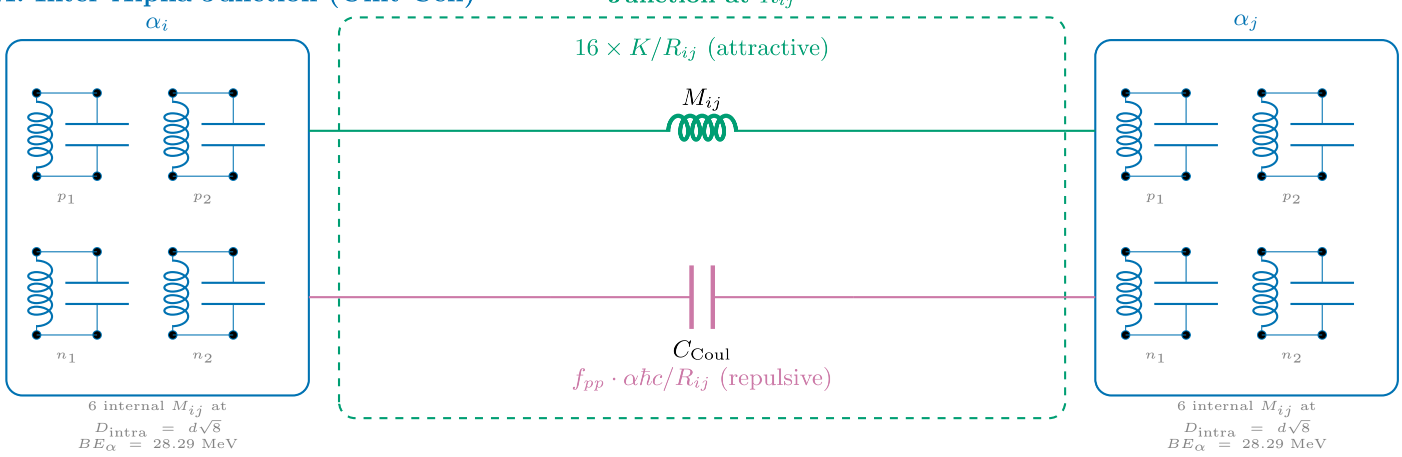
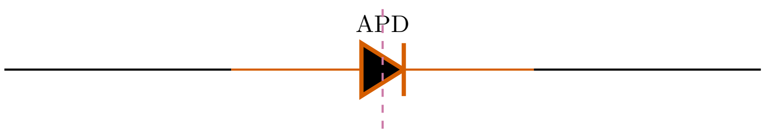
F-19 Semiconductor Equivalent Circuit

$4\alpha +^3$ H Core+Halo — $V_R/V_{BR} = 0.030$ — $M = 1.000$ (Core+Halo)

A: Inter-Alpha Junction (Unit Cell)



B: Miller Avalanche Stage

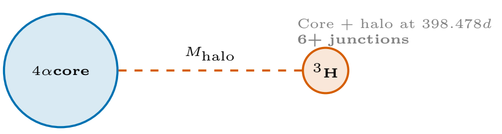


$V_{BR} = 6\alpha\hbar c/D_{\text{intra}} = 3.594$ MeV

$$\boxed{M = \frac{1}{1 - \left(\frac{V_R}{V_{BR}}\right)^5}}$$

F-19: Core $V_R/V_{BR} = 0.030 \implies M = 1.0$ (Core inherits parent regime)

C: F-19 Topology (6 junctions)



D: Energy Balance

$$M_{\text{nuc}} = 4\,M_\alpha - BE_{\text{net}}$$

$$BE_{\text{net}} = \underbrace{\sum_{i<j}^{6} 16 \cdot \frac{K}{R_{ij}}}_{textStrong(attractive)} - \underbrace{M \cdot \sum f_{pp}\,\frac{\alpha\hbar c}{R_{ij}}}_{\text{Coulomb (repulsive)}}$$

Strong: $\sum K/R =$ engine-derived

Coulomb: $\sum \alpha\hbar c/R \times f_{pp}$

Miller: $M = 1.000$ ($V_R/V_{BR} = 0.030$)

Result: $17,692.302$ MeV (0.0000% error)

Each $\alpha = 2p + 2n$ LC tanks at tetrahedron vertices ($D_{\text{intra}} = d\sqrt{8} \approx 2.38$ fm). $d = 4\hbar/m_p c \approx 0.841$ fm. $K_{\text{mutual}} = 11.34$ MeV·fm. $V_{BR} = 3.594$ MeV.